

Experiment 5

Develop an interactive image classification application using a trained deep-learning model. The application should allow a user to upload an image, display the model's predictions and confidence scores, and use Integrated Gradients to visualize the image regions contributing to a selected prediction. Analyze whether the model is focusing on relevant regions of the image and investigate the effect of different baseline inputs on the generated explanation.

Minimum Requirements

Your application must:

1. Provide an interactive user interface.
2. Allow a user to provide an image to the application.
3. Use a 'ResNet18 pretrained on ImageNet-1K' image classification model at the backend.
4. Generate and display the model prediction.
5. Apply Integrated Gradients to explain the model's prediction.
6. Provide an appropriate visual representation of the regions contributing to the model's decision.
7. Work with different compatible images and not only with a single predefined image.